

GFPO baseline (Shrivastava et al., 2025). GFPO oversamples G responses, keeps a top- k subset S , and zeroes gradient from the dropped responses.

$$S = \text{TopK}_m\{1, \dots, G\},$$

$$m_i = \begin{cases} -L_i, & \text{shortest filter} \\ r_i / \max(L_i, 1), & \text{token-efficiency filter} \end{cases}$$

$$\mu_S = \frac{1}{|S|} \sum_{j \in S} r_j, \quad \sigma_S = \sqrt{\frac{1}{|S|} \sum_{j \in S} (r_j - \mu_S)^2}$$

$$A_i^{\text{GFPO}} = \mathbf{1}[i \in S] \frac{r_i - \mu_S}{\sigma_S + \epsilon}.$$

Our reproduction uses $G = 16$, $k = 8$, raw rewards unchanged, and dropped responses kept in batch with zero advantage.